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Bengali Hate Speech Categorization: A Comparison of Hybrid Deep Learning and Transformer-Based Models

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***Abstract*—The proliferation of hate speech in online social engagement platforms poses significant challenges to digital communication platforms, particularly in linguistically diverse regions such as South Asia. The rapid growth of virtual entertainment has facilitated easier interactions across online platforms, but this has also led to an increase in targeted hate speech. This paper presents a system for categorizing hate speech in Bengali text, leveraging a combination of hybrid deep learning and transform learning techniques. We constructed a dataset of approximately 19,398 Bengali texts from YouTube comments, which were manually classified into six categories: Hate Speech, Religious Hatred, Inciteful, Political Comments, Religious Comments, and Communal Attack. Among hybrid deep learning models, CNN-BiLSTM achieved the highest accuracy of 76%, surpassing both the GRU-LSTM and BiGRU-CNN models. Notably, BanglaBERT excelled as the top-performing Transformer algorithm with 80% accuracy. This exhibits the efficacy of the hybrid model in categorizing hate speech within Bengali text. We have also evaluated the effectiveness of our proposed approach by comparing with baseline methods.**

***Index Terms*—Hate Speech, Hybrid Deep Learning, BanglaBERT, CNN-BiLSTM, GRU-LSTM, BiGRU-CNN.**

I. Introduction

People's everyday lives are increasingly integrating social media platforms into their daily activities. These platforms have made information more accessible to everyone, giving people greater autonomy of speech, and occasionally even providing concealment. Facebook continues to be one of the most widely used social engagement sites worldwide and a major platform for private communication, with approximately 2.9 billion active users per month [1]. As of April 2024, Bangladesh had approximately 55.7 million Facebook users, ranking it among the top countries by Facebook audience size [2].

The issue of hate speech has become increasingly pressing, primarily due to the widespread adoption of social networking sites and the internet. Hate speech and deceptive data are being freely disseminated on these websites. The vast volumes of textual data generated by users on social media through posts, comments, and messages have spurred significant research interest in understanding and addressing this issue. As the process of detecting hate speech online has become more automated, interest in its impact has surged. There are various challenges in identifying hate speech on social media networks, such as limited and imbalanced datasets, selecting a suitable model, and determining the right feature analysis technique.

Since the advent of artificial intelligence, numerous approaches have been explored to tackle various problems. Learning algorithms, such as deep learning and machine learning have gained significant popularity and are widely applied to address classification and recognition challenges. Artificial neural networks inspired by the human brain are a key component of hybrid deep learning in artificial intelligence. These models strategically combine different neural network architectures to harness their specific advantages. Transformer based text detection is a specialized subset of equipment nameplate text detection. In natural language processing, the Transformer stands out for its ability to manage long-range dependencies and efficiently tackle sequence-to-sequence tasks. BERT, one of the most widely used Transformer models, is highly regarded for its ability to process text across multiple languages.

This research seeks to bridge gaps in existing studies by developing a model capable of accurately categorizing hate speech in Bengali comments. Using transformer-based algorithms, hybrid deep learning, and various NLP techniques,

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the study makes the following significant contributions:

- Creation of a new Bengali dataset comprising of 19,398 texts from YouTube comments.
- Comparative analysis among the algorithms used to determine the effectiveness of the proposed system.
- The proposed approach is benchmarked against prior methods, offering a more robust model.

II. RELATED WORKS

This section emphasizes the significance of recent survey and review articles, particularly highlighting the valuable contributions they make to the field. Here an overview of the latest studies has been presented on hate speech detection and classification.

Ishmam et al. conducted a study using machine learning for identification of hate speech in Bengali Facebook comments, categorizing them into six distinct groups [3]. They employed deep neural networks, machine learning algorithms, and gated recurrent units (GRUs) for this task. Their GRU-based models achieved a notable accuracy of 70%, outperforming Random Forest models by approximately 18%. While their results were impressive, our model surpassed theirs, achieving an 80% accuracy rate. Moreover, a deep learning method for identifying hate speech in Bangla text on social media is put forth by Das et al. [4]. Facebook page comments are analyzed and classified into seven types. CNN+LSTM, CNN+GRU, and CNN+Attention are some of the combinations that are evaluated. According to their research, the attention-based decoder had the best multi-class hate speech identification accuracy (77%) out of all of them.

A Bangla explainable hate speech detection system was presented by Karim et al. [5]. Their DeepHateExplainer system efficiently identifies hate speech in Bengali with fewer assets. The model was created by the researchers utilizing CNN-BiLSTM, XLM-RoBERTa, multilingual BERT, and BanglaBERT after they gathered 5700 Bangla text data. This dataset was comparatively smaller than the one that was used in this research. Moreover, Ghosh et al. [6] achieved 84% accuracy in separating hostile Bengali texts by combining machine learning and deep learning. A dataset with fewer classes was used. Additionally, deep learning and machine learning were used by Shrivastava et al. [7] to identify hate speech in Bengali. Their best accuracy of 65% indicates that there is still room for improvement.

Romim et al. [8] introduced the substantial dataset BD-SHS with the aim of identifying hate speech in Bangla. They provided BD-SHS, a large, manually labeled dataset that covers hate speech in a variety of social settings, and draw attention to the shortcomings in the existing Bangla hate speech datasets. The dataset is at least 60% larger than any other Bangla HS dataset currently available and contains over 50,200 abusive remarks that were scraped from social networking sites online. The best F1-score with many NLP models is 91.0%. This dataset and methodology advance the identification of hate speech in Bangla while offering valuable resources for further study. Abdhullah-Al-Mamun et al. [9] collected 2400 Facebook and Twitter public status updates in Bangla from and used the SVM, J48, KNN, and Naïve Bayes algorithms in two stages. In the initial phase, they trained with just the texts. In the second stage, some user data and texts were added, and the models were retrained. SVM again achieved the best accuracy of 97.27%. The models with the highest accuracy of 95.40% were those that used SVM.

In their study [10], Luo et al. developed a multi-featured cyberbullying detection system based on deep learning methods using a BiGRU-CNN architecture. They collected data from twitter. The study concentrated on using deep learning techniques to identify cases of cyberbullying and claimed to have achieved an 89% accuracy rate with their proposed model. After gathering Bangla texts with emoticons and Unicode Bangla letters, Chakraborty et al. [11] used LSTM, SVM, CNN, and MNB, with SVM exhibiting the best performance with 78% accuracy. Similar to this, Ahammed et al. [12] obtained a maximum accuracy of 72% by gathering Bangla texts from Facebook and applying SVM and Naïve Bayes.

III. METHODOLOGY

A. Workflow

This research presents a hate speech categorization system for Bengali texts. For that purpose, a dataset was constructed containing 19,398 Bengali texts in six categories. For classification, hybrid deep learning and transform learning-based approaches are tested, and finally, our model is also benchmarked against existing literature to further demonstrate its efficiency. Fig. 1 demonstrates our proposed methodology.

B. Dataset

In this work, three data sets have been utilized. The descriptions of these datasets are as follows:

The first dataset, which we collected from Github [13], was utilized by Ishmam et al. in [3]. This dataset contains 5,126 texts that are labeled into six categories: Hate Speech, Religious Hatred, Inciteful, Political Comments, Religious Comments, and Communal Attack. For the rest of this paper, this dataset will be referred to as Dataset I. We first used this dataset to train our algorithms and tested their accuracy to see if the models could surpass the existing accuracy.

We built another dataset by collecting data from YouTube using the YouTube API. We collected information from the comment sections of YouTube videos featuring numerous online entertainers from Bangladesh to create this dataset. 19,398 Bengali comments in all were acquired. The dataset was manually labeled into six categories: Hate Speech, Religious Hatred, Inciteful, Political Comments, Religious Comments, and Communal Attack. For the rest of this paper, this dataset will be referred to as Dataset II.

Our third dataset is a combination of Dataset I and Dataset II and consists of 24,524 labeled Bengali texts divided into the same six categories. For the rest of this paper, this dataset will be referred to as Dataset III.

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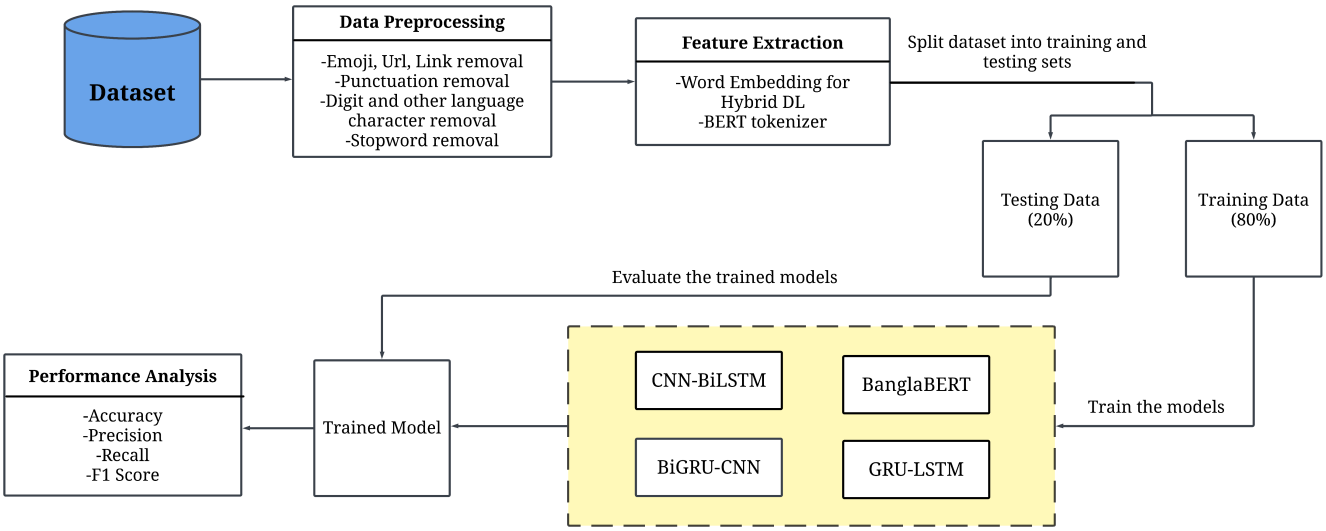

Fig. 1: Proposed Methodology

C. Preprocessing

Due to the significant noise in the datasets, it could disrupt the classification process. Therefore, cleaning the dataset is necessary to enhance the model's performance. We removed punctuation marks, English characters, numerals, emoticons, links, URLs, special characters, user tags, and mentions from all three datasets. Additionally, data consisting of only a single word in a sentence was eliminated.

D. Feature Extraction

1) Word Embedding : In this study, a crucial component of our hybrid deep learning model for text classification was the embedding layer. This layer transforms input text into dense numerical vectors of a fixed size, allowing the model to effectively learn semantic word representations from the dataset. Our objective is to improve the model's understanding of text by using an embedding layer to capture the semantic relationships between words.

2) BERT Tokenizer: BERT models include a built-in tokenizer, removing the need for extra word embedding techniques. This tokenizer produces three arrays:token type IDs, input IDs, and attention masks. In this study, we will concentrate on using the token type IDs and input IDs.

E. Splitting the dataset into train and test data

The dataset is splitted into training and testing sets following tokenization and word embedding. Eighty percent of the data, or the training set, is used to train the algorithms, while the remaining twenty percent is utilized to assess performance. This ratio of data splits has been kept constant. The sizes of the training and testing datasets are shown in Table I.

TABLE I: DATASET DESCRIPTION AFTER TRAIN TEST SPLIT

Datasets	*Train Data*	Test Data
Dataset I	4100	1026
Dataset II	15518	3880
Dataset III	19619	4905

F. Hybrid Deep Learning Algorithms

In this study, deep learning algorithms were used in a hybrid approach. The details of the algorithms are presented here.

1) CNN-BiLSTM: The CNN-BiLSTM model integrated bidirectional long short-term memory networks (BiLSTMs) [14] with convolutional neural networks (CNNs) [15] to form a hybrid architecture. Key components of this model include Embedding Layer, Conv1D Layer Bidirectional LSTM Layer, Max-Pooling Layer, Flatten Layer, Dropout Layer, Dense Layer and Regularized Dense Layer.

2) GRU-LSTM: The GRU-LSTM stands for Gated Recurrent Units (GRUs) and Long Short-Term Memory (LSTM) combines the layers for text classification. GRUs, which are quicker to train and capture sequential dependencies with fewer parameters [16], work alongside LSTMs that retain long-term dependencies [17]. Dropout layers are used to prevent overfitting.

3) BiGRU-CNN: Convolutional neural networks (CNNs) and bidirectional GRUs (BiGRUs) combine in a BiGRU-CNN text categorization model. The BiGRU layer helps grasp dependencies by storing context in the text sequence from both directions. Convolutional filters are utilized by the Conv1D layer to derive local features from the sequences. To be able to ensure broadness and avoid overfitting, dropout layers

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are employed. Lastly, the class values for classification are generated by dense layers using a softmax activation function.

G. Transformer Based Algorithm

BanglaBert: The development of pre-trained language models with domain-specific BERT-like features is gaining attention. Similarly, BanglaBERT [18] has been promoted and demonstrated to perform better than previous BERT models for Bengali text classification. BanglaBERT is a transformer-based language model designed to effectively understand and generate text that is spoken only in Bengali. It leverages the powerful contextual understanding of language provided by the BERT architecture, which has raised the bar for natural language processing (NLP).

H. Performance Analysis

As the models were trained, we have assessed the performance in terms of accuracy, precision, recall, and f1-score. Additionally, accuracy and loss curves are used to evaluate the performance of the model.

IV. RESULTS AND DISCUSSION

In this section, we wander over each model's performance on all three datasets used in this research.

Table II depicts the precision, recall, and f1- score values of the 4 algorithms for Dataset I.

TABLE II: PERFORMANCE OF HATE SPEECH CATEGORIZATION FOR DATASET I

Classifier	Precision	Recall	F1-score	Accuracy
CNN-BiLSTM	0.76	0.76	0.76	0.76
BiGRU-CNN	0.76	0.76	0.76	0.76
GRU-LSTM	0.74	0.74	0.74	0.73
BanglaBERT	**0.79**	**0.80**	**0.80**	**0.80**

Table II depicts that, with an average precision score of 0.79, recall values of 0.80, and F1 scores of 0.80 of all classes, BanglaBERT was the best-performing model for dataset I. BanglaBERT obtained 80% overall accuracy. GRU-LSTM, on the other hand, produced the worst results, averaging 0.74 for precision across all classes and 0.74 for recall and F1-score. Its accuracy score was 73%.

As this dataset was used by Ishmam et al. [3], we have presented a comparison in Table III between their approach and our approach.

TABLE III: COMPARISON WITH PREVIOUS LITERATURE

Approach	Algorithm	Precision	Recall	F1-score	Accuracy
Previous Approach [3]	GRU	0.68	0.70	0.69	0.70
	Random Forest	0.56	0.52	0.5	0.52
Our Approach	CNN-BiLSTM	0.76	0.76	0.76	0.76
	BiGRU-CNN	0.76	0.76	0.76	0.76
	GRU-LSTM	0.74	0.74	0.74	0.73
	BanglaBERT	0.79	0.80	0.80	0.80

Our model achieved 80% accuracy with BanglaBERT, which is better than the prior method, as shown in Table III. Accuracy rates for previous approaches, like Random Forest and GRU, were 52% and 70%, respectively. With BanglaBERT attaining an accuracy of 80%,CNN- BiLSTM and BiGRU-CNN at 76%, and GRU-LSTM at 73.58%, our method clearly shows considerable gains. These findings demonstrate how well our suggested techniques—in particular, the BanglaBERT model—work for precisely categorizing hate speech in Bangla text.

Furthermore, we analyzed the performance of our methods for Dataset II and Dataset III.

Table IV presents the precision, recall, and f1- score values of the 4 algorithms for Dataset II.

TABLE IV: PERFORMANCE OF HATE SPEECH CATEGORIZATION FOR DATASET II

Classifier	Precision	Recall	F1-score	Accuracy
CNN-BiLSTM	0.86	0.86	0.86	0.86
BiGRU-CNN	0.84	0.84	0.84	0.84
GRU-LSTM	0.85	0.85	0.85	0.85
BanglaBERT	**0.92**	**0.92**	**0.92**	**0.92**

BanglaBERT was the most accurate model for dataset II, averaging 0.92 precision scores for all classes. Additionally, it obtained the same f1-score and recall in every class. BanglaBERT obtained a 92% accuracy. On the other hand, with average recall scores of 0.84, average F1-scores of 0.84, and average precision scores of 0.84 across all classes, BiGRU-CNN exhibited the lowest scores. It obtained an accuracy score of 84%.

Table V depicts the precision, recall, and f1- score values of the 4 algorithms for Dataset III.

TABLE V: PERFORMANCE OF HATE SPEECH CATEGORIZATION FOR DATASET III

Classifier	Precision	Recall	F1-score	Accuracy
CNN-BiLSTM	0.76	0.76	0.76	0.78
BiGRU-CNN	0.76	0.76	0.76	0.79
GRU-LSTM	0.74	0.74	0.74	0.78
BanglaBERT	**0.79**	**0.80**	**0.80**	**0.84**

The highest performance for Dataset III was again shown by BanglaBERT. It obtained 0.79 precision, 0.80 recall, and 0.80 F1-scores on average across all classes. The accuracy score obtained by BanglaBERT for Dataset III is 84%. Conversely, GRU-LSTM failed to perform admirably in this case, with an overall accuracy of 78%. Fig. 2 presents a comparison of the accuracy scores of all the algorithms, across all datasets.

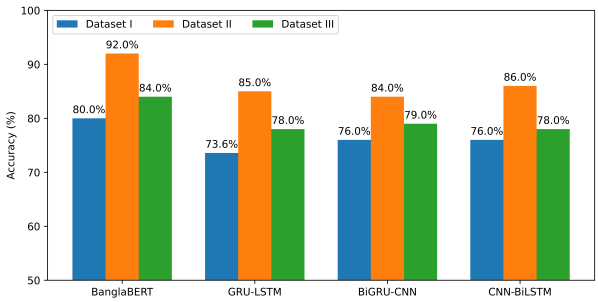

Fig. 2: Accuracy chart for all datasets

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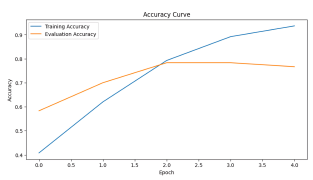
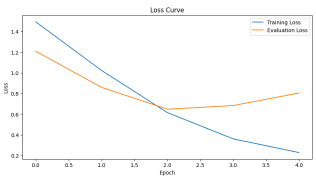
The accuracy chart for the algorithms is displayed in Fig. 2, which facilitates a clear visual depiction of how each algorithm performs in various scenarios by allowing one to compare its performance across the three datasets. BanglaBERT had the greatest accuracy out of every technique, obtaining 80% for Dataset I, 92% for Dataset II, and 83% for Dataset III. Two of the three hybrid deep learning models, CNN-BiLSTM and BiGRU-CNN, both performed well, achieving an accuracy of 76% for Dataset I. To be exact, CNN-BiLSTM outperformed BiGRU-CNN for Dataset III, with an accuracy of 79%, whereas CNN-BiLSTM was better for Dataset II.

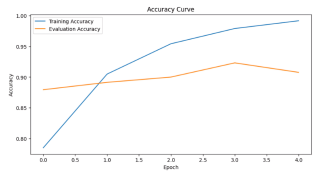
Additionally, Fig. 3 provides visual representations of the training-validation accuracy and loss over epochs graph to provide deeper insights into our best-performing algorithm.



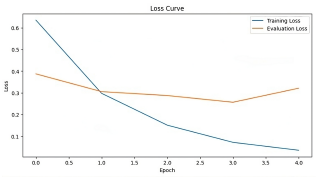
(a) Epoch vs Accuracy (Dataset I)



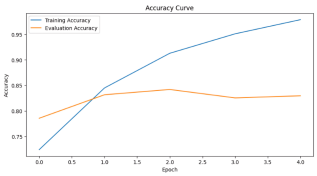
(b) Epoch vs Loss (Dataset I)



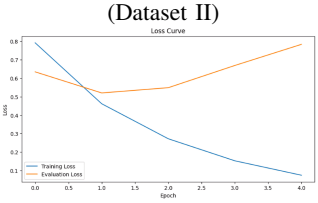
(c) Epoch vs Accuracy (Dataset II)



(d) Epoch vs Loss (Dataset II)



(e) Epoch vs Accuracy (Dataset III)



(f) Epoch vs Loss (Dataset III)

Fig. 3: Training and validation accuracy and loss over epoch for BanglaBERT

As shown in Fig. 3, for BanglaBERT, the training accuracy steadily increased over the epochs, signaling effective learning. The validation accuracy has also shown an almost steady increase for all three datasets. The training loss declined consistently, which is a positive sign of model convergence. On the contrary, the validation loss started to increase after a certain point for dataset III, but showed an almost steady decrease for Dataset I and II.

V. CONCLUSION AND FUTURE WORK

Our research presents a significant advancement in hate speech categorization in the field of Bengali language. By leveraging a meticulously curated dataset of 19,398 YouTube comments and employing rigorous data cleaning techniques, we have conducted a research that surpasses previous approaches in accuracy. Among the deep learning algorithms tested, CNN-BiLSTM emerged as the front runner, achieving an impressive 76% accuracy rate. This outperformed for GRU-LSTM and BiGRU-CNN, which scored 73.58% and 76%, respectively. Nonetheless, the Transformer-based BanglaBERT exhibited superlative performance, attaining a remarkable 80% accuracy. This research provides a valuable benchmark for future studies in hate speech detection, categorization and related areas, offering insights into the effectiveness of different deep learning architectures and the importance of robust data preparation. Future aspects of this research include benchmarking the Dataset II and experimenting with additional transformer-based algorithm variations.

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